ModelIt!

Student Activity Pack

When Permafrost Melts, What Wakes Up?

Modeling Arctic Carbon Release and Climate Amplification

Grade 11L2 | NGSS HS-ESS2-4, HS-ESS2-6 | Cause and Effect / Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think is frozen beneath the surface of the Arctic, and how long has it been there?

2 Why might the Arctic be warming faster than the rest of the planet?

3 Draw a diagram showing what you think happens when frozen ground thaws — what is released and where does it go?

4 What is a positive feedback loop, and can you think of an example in nature?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Arctic Temperature Anomaly
- Permafrost Thaw Depth
- Carbon Release Rate
- Atmospheric Greenhouse Gas Level
- Global Warming Rate
- Thermokarst Formation Rate

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate Warming

+2 degrees C Arctic warming, gradual thaw

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Trajectory

+4 degrees C Arctic warming, accelerating thaw

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Runaway Feedback

+6 degrees C Arctic, thermokarst collapse

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Abrupt Methane Release — Sudden, explosive releases of methane from thawing clathrate deposits or pressurized gas pockets beneath permafrost — these events create craters tens of meters wide and release massive quantities of greenhouse gas in seconds

Albedo Change — As permafrost landscapes shift from snow-covered tundra to dark lakes and bare soil, they absorb more solar radiation instead of reflecting it, creating an additional warming feedback that accelerates local temperature increase

Microbial Community Shift — As permafrost thaws, microbial communities shift from dormant cold-adapted organisms to active decomposers that metabolize the ancient organic carbon — the composition of these communities determines whether carbon is released as CO_2 or the more potent methane

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens when 1.5 trillion tons of frozen carbon — twice as much as exists in the entire atmosphere — starts thawing and releasing greenhouse gases? Use evidence from your simulation to support your answers.

1 Why is the permafrost carbon feedback fundamentally different from fossil fuel emissions in terms of human ability to control it?

2 How does your model demonstrate that Arctic warming disproportionately affects global climate even though the Arctic is sparsely populated?

3 What does the positive feedback loop in your model suggest about the urgency of emission reductions relative to permafrost tipping points?

4 How would thermokarst formation change the predictions of your model compared to gradual surface thaw alone?

5 What does your model reveal about the difference between slow, manageable permafrost thaw and rapid, irreversible carbon release?

8 STEM Challenge: Design a Permafrost Carbon Monitoring Network

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YOUR ENGINEERING MISSION

Design a comprehensive monitoring system for an Arctic region that can track permafrost thaw progression, quantify carbon release rates, and provide early warning of runaway feedback conditions.

THE SCENARIO

A consortium of Arctic nations has funded a monitoring project to track permafrost thaw across a 10,000 square kilometer region in Siberia. Your team must design a sensor network that combines ground-based measurements, satellite remote sensing, and atmospheric sampling to quantify carbon release and detect acceleration in the feedback loop.

Engineering Design Thinking

1. WHAT TYPES OF SENSORS (TEMPERATURE, GAS FLUX, GROUND PENETRATING RADAR) WOULD YOU DEPLOY AND AT WHAT DENSITY?
2. HOW WOULD YOU DESIGN ATMOSPHERIC SAMPLING TO DISTINGUISH PERMAFROST METHANE FROM OTHER SOURCES?
3. WHAT SATELLITE IMAGERY ANALYSIS WOULD DETECT THERMOKARST LAKE FORMATION AND EXPANSION OVER TIME?
4. HOW WOULD YOU INTEGRATE YOUR DATA INTO EXISTING GLOBAL CLIMATE MODELS TO IMPROVE PERMAFROST CARBON FEEDBACK PROJECTIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Permafrost Scientists and Cryosphere Researchers study frozen ground systems and their response to climate change, deploying monitoring equipment in some of Earth's most remote and harsh environments. They work for research institutions like the Alfred Wegener Institute, University of Alaska Fairbanks, and the Geological Survey of Canada, earning $70,000–$135,000/year. Climate Modelers specializing in carbon cycle feedbacks earn $85,000–$155,000/year.